Network Swiss Army Knife (NSAK)

Bachelor Thesis – Integrating Agentic AI for Network Reconnaissance

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Overview

- Introduction
- Implementation
- Evaluation Setup
- Ai Integration
- Results
- Discussion and Conclusion
- Demo: Agentic Reconnaissance in Action

Introduction

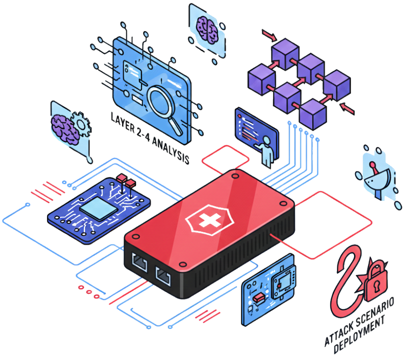
Starting Point: The NSAK Framework

Established in the predecessor project (Project 2)

A modular, open-source network security framework

Built around four reusable resource types:

- **Device** *execution host at a network position*
- **Environment** *target network topology*
- **Scenario** *a red/blue team use case*
- **Drill** *reusable building block*



- Gather baseline data
- Detect
- Alert
- Investigate
- Plan a response
- Execute

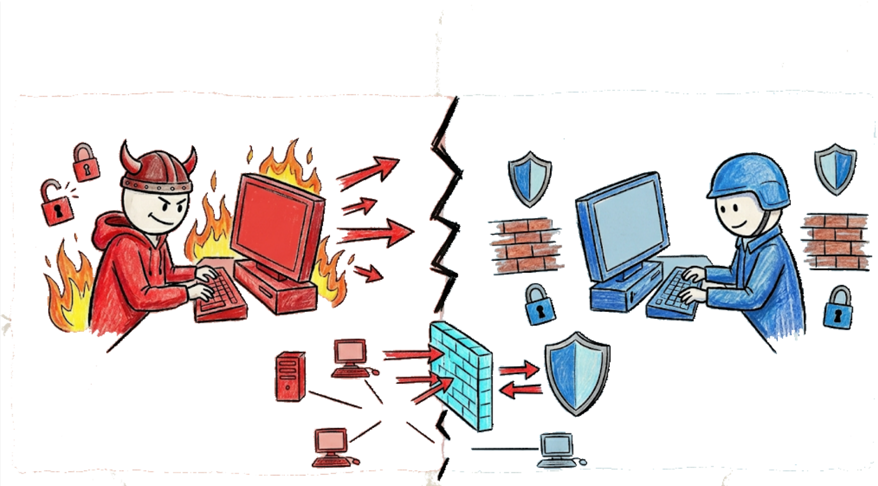


Figure: Red and Blue Team

Reconnaissance

The first stage of the cyber kill chain

Gathering information about a target network *before* any attack – it builds the map every later stage relies on, and is the focus of this thesis.

- **Host discovery** *which systems are reachable on the network*
- **Service & port enumeration** *what is running and exposed*
- **Characterization** *OS, versions and vulnerability indicators*

Hypothesis and Research Questions

From static scenarios to agentic AI

Hypothesis

Integrating **agentic AI** into NSAK improves **adaptability, flexibility** and **automation** over static scenarios, while **lowering the operator's required expertise** to interpret results and plan follow-up steps.

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RQ1 How does an agentic AI scenario perform **compared to a static reference** scenario in network reconnaissance?

RQ2 To what extent can agentic AI **support operators** during interactive reconnaissance?

RQ3 To what extent is the implementation **suitable for real-world deployment**?

Project Goals

Order of implementation: Must → Should → Could

Must

- Compare Frameworks
- AI Integration Research
- Configuration Management
- AI Scenarios
- Reproducible Test Env.
- Red Team Reconnaissance
- AI Scenario Evaluation
- AI vs. Engineered (Red)

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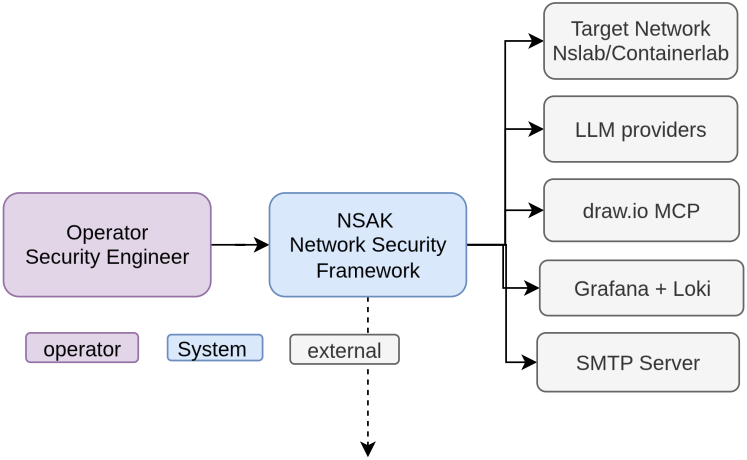
REST API
MCP Server
Web GUI

Implementation

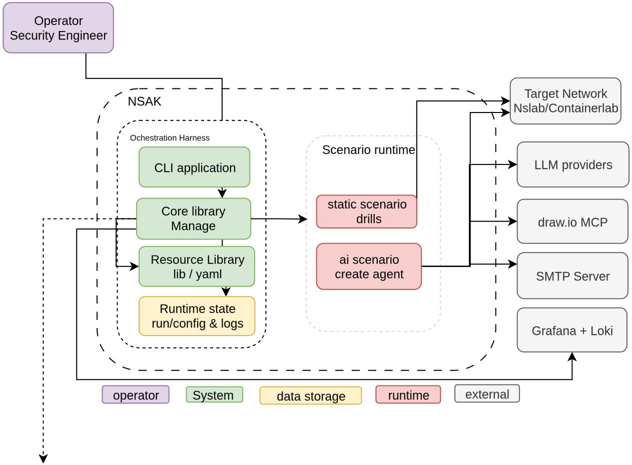
Layered Architecture

system-diagram



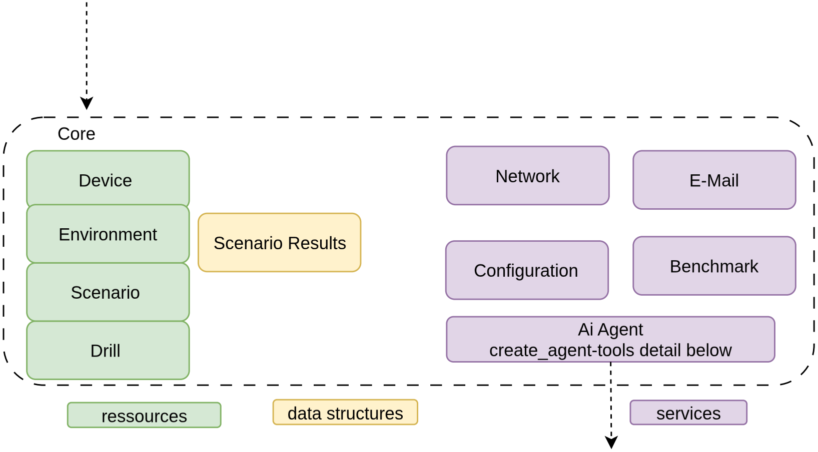
Layered Architecture

Lean Component Diagram



Layered Architecture

Core Diagram and Agent



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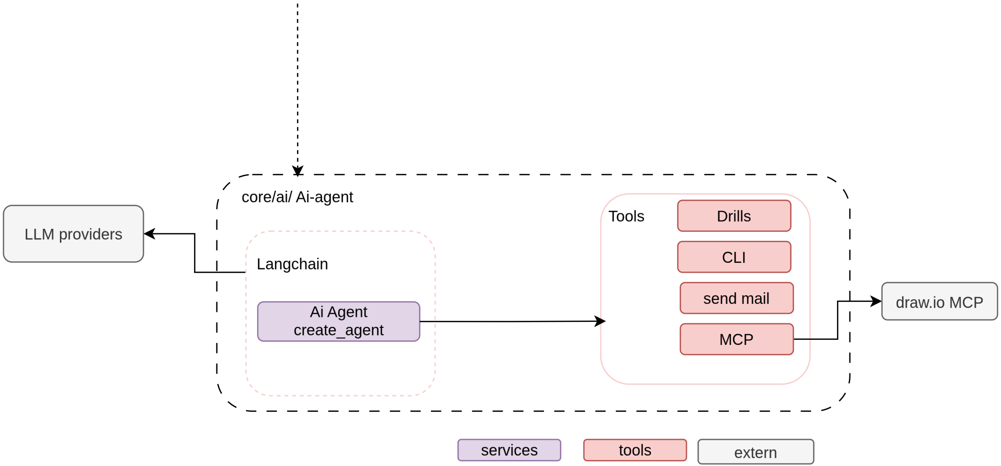


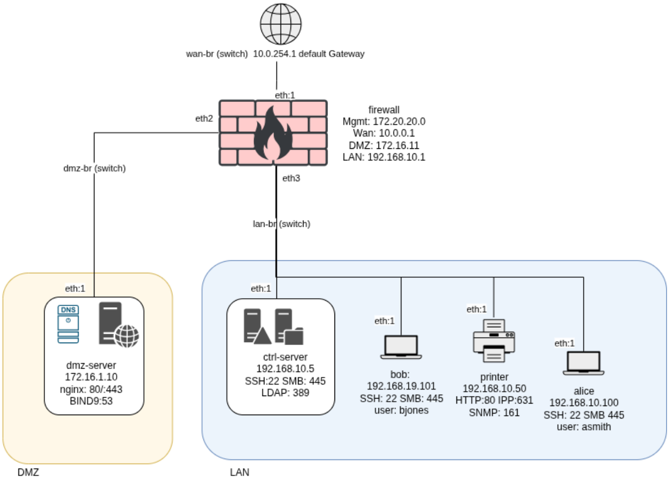
Evaluation Setup

Methology

3 LLMs • 2 Network Environments • Static Reference Baseline

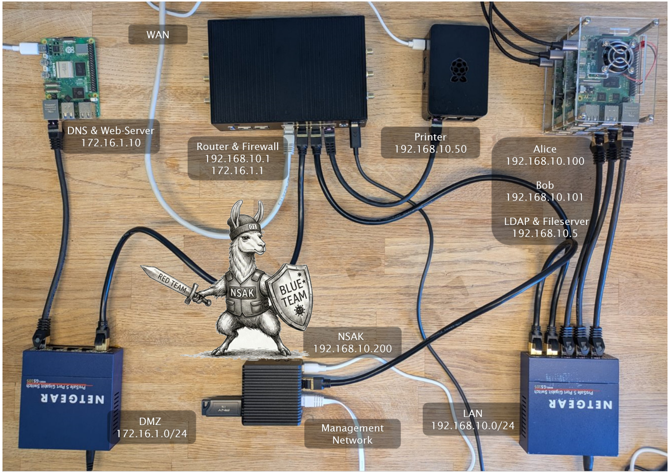
Reproducible Test Environment

A virtual enterprise network built with Containerlab



Physical Lab

The same network implemented with real hardware



Benchmarking Suite

Automated test runs for non-interactive scenarios

Ai Integration

1. Provider Model und Agent
2. Tool calling (Langchain Tools) (MCP) (human interaction hook) (cli) (draw io) (send-mail)
3. 1. Single Agent interactive unstructured output
4. 2. Single Agent Reconnaissance unstructured output
5. 3. POC Blueteam intrusion detection - not followed time and scope - unstructured output
6. 4. Single Agent reconnaissance structure output (context-window, grammar -> why structured output create larger grammar)
7. 5. Multi Agent reconnaissance structured output (issue with smaller models) () Containerlab evalutation smaller model
8. 6. Multi Agent reconnaissance unstructured output

Concept 1: Taming the Context

From one overloaded agent to a focused pipeline

⚠ The problem

- One agent does everything in a single long conversation
- Every scan output piles up → the **context keeps growing**
- Smaller models lose the thread and start failing
- Cost and runtime become hard to predict

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- Split the task into **discovery** → **enumeration** → **assessment**
- Each phase is a **fresh agent** with its own, small context
- We pass on only the **distilled result**, not the whole history

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Trade-off: rescues small models, but adds orchestration overhead that frontier models don't need.

Concept 2: Keeping the Operator in Control

The agent proposes – the human decides on risky actions

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- The agent **pauses** before sensitive actions and asks for a decision
- Shell commands → **approve** before they run
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After the decision the agent resumes exactly where it stopped – no restart, no lost progress.

Single Agent

- One prompt to achieve the goal
- Long chain of tool calls, **growing context**
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Multi Agent *divide-and-conquer*

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- Smaller, focused context per step
- Helps small models; may add overhead for frontier

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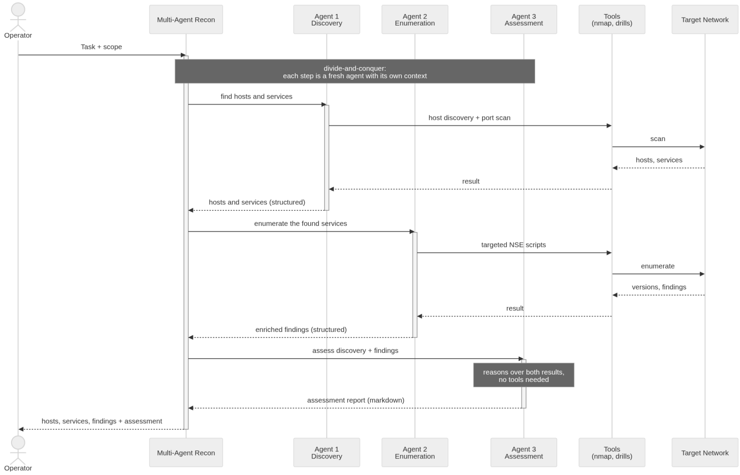
Structured vs. Unstructured output

Structured (schema-constrained): great for automated metrics, but **high failure rate** – especially for small models.

Unstructured: needed in the BFH lab where all models failed to produce valid structured output; freed the agent but required manual evaluation.

Multi-Agent Reconnaissance Pipeline

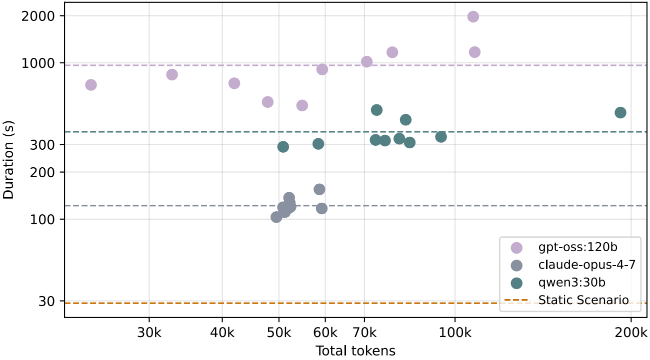
Discovery → enumeration → assessment, each a fresh agent



Results

Containerlab – Multi-Agent Benchmark

Containerlab Findings



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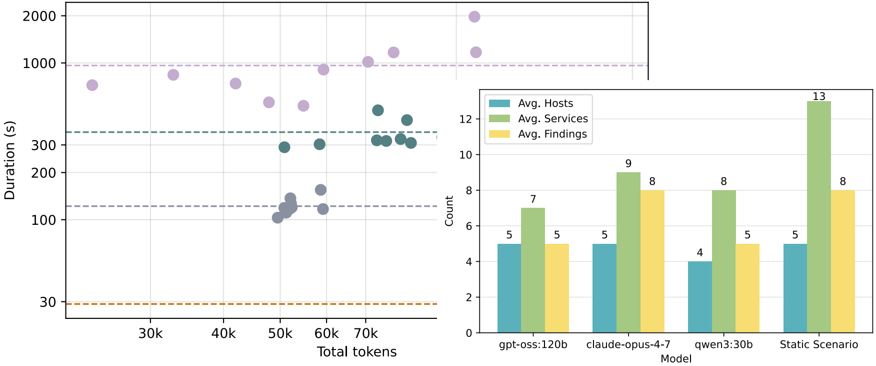

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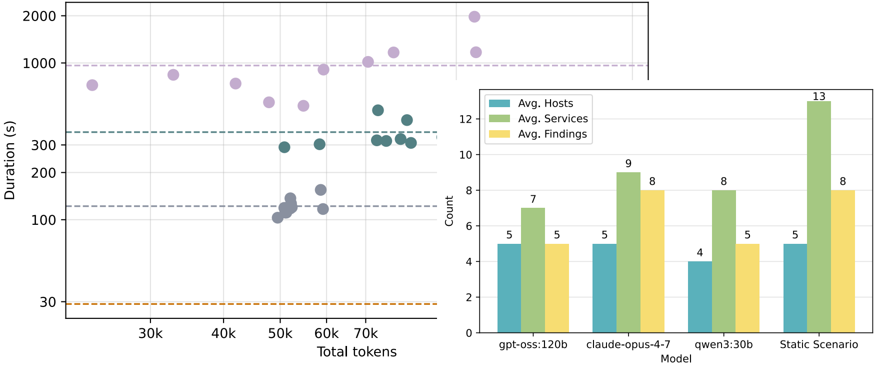


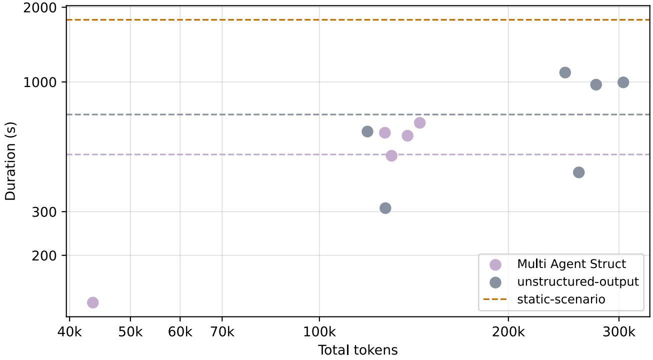
- Decomposition: `qwen3:30b` drops tokens **−68%** and duration **−30%**.
- `claude-opus-4-7` keeps identical discovery quality (slight +13% token overhead).

single agent resultate -> Lösung zu Multiagent Structured vs Unstructured interactive

Qualitative + Interactive

Network Security Lab – Multi-Agent Benchmark

NSlab - Findings



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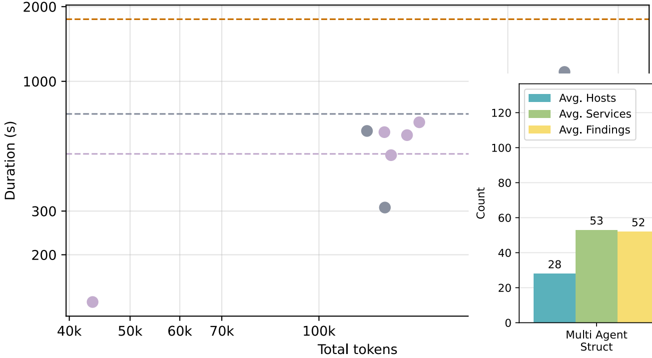

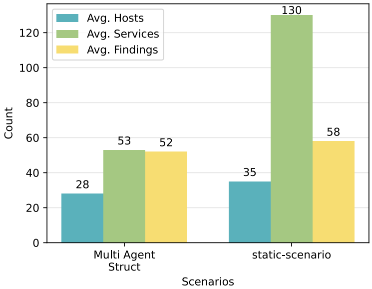

Discussion and Conclusion

Key Findings

Takeaways across both environments

What works

- Even smaller models can autonomously run **standard red-team reconnaissance**
- **Multi-agent decomposition** helps smaller models (–68% tokens for `qwen3:30b`)
- AI delivers **depth**: assessment, prioritized findings, remediation

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What limits it

- **High failure rate** – best-case subset
- **Hallucinations**: fabricated CVEs, honeypots

Answering the Research Questions

RQ1 **AI vs. static:** Same discovery power as baseline, no gains in speed/determinism. Adds value via **reasoning and interpretation** (credentials, banner reasoning, severity scoring).

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RQ2 **Operator support:** Converts scans into structured, severity-ranked output + next steps. **`--interactive`** Operator stays in the loop and control.

RQ3 **Deployment:** Scales to ∼35 hosts, highlights key risks. **Requires supervision** (reliability limits).

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Recommendations and Future Work

Recommendations

- **Match model tier to task** – frontier model for autonomous recon
- **Multi-agent** for smaller / self-hosted models
- **Hybrid pipeline**: static breadth + AI depth (+ RAG)
- **Human-in-the-loop** + retry on incomplete runs

Future work

- **Dynamic tool calling** – expose only relevant tools
- **Guardrails & sandboxing** – scope restriction for production
- Robust **HITL middleware**
- **Broader attack-chain** coverage beyond reconnaissance

Conclusion

Agentic AI complements deterministic enumeration

- ✔ Using NSAK directly as an **agentic harness** proved effective – the agent autonomously and interactively executes network discovery via LangChain tools.
- ✔ Scripted scenarios excel in correctness and completeness; the agent adds **reasoning and assessment**, limited by model capability.
- ✔ **Multi-agent decomposition** models to digest large context improves models by reducing context complexity and increasing consistency.
- ⚠ Hallucinations remain a limitation – **supervised operation with a human reviewer** is the proposed deployment model.

LLMs make cybersecurity use cases more accessible and cost-effective – with a human in the loop.

Personal Conclusion

Frank

-

Lukas

- Initially skeptical of the usefulness of AI
- We covered a lot of topics we learned at BFH
- A bit of naivety in the planning led to a larger scope
- Keep focusing on the goals and research questions
- AI is here to stay as another engineering tool

Demo: Agentic Reconnaissance in Action